MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
TYPES 12AVP7, 12AVP19

The complete requirements for procuring the products described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus.

BASE: B12-37 OUTLINE, DIMENSIONS, PIN CONNECTIONS: See Fig. 1 herein.

RATINGS (ABSOLUTE): (Note 1)

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Eb3	Rg	Zd	Ehk	Eb3/Eb2	Alt.
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc	Meg	Meg	Vdc	Ratio	ft
Maximum:	6.3 ± 10%	0	1500	3000	6700	11,000	2.0	5.0	±180	2.0	10,000
Minimum:	---	-300	---	---	---	---	---	(Note2)	---	---	---
TEST CONDITIONS: 6.3	Adjust	---	Focus	6100	9700	---	---	---	---	---	---

MIL-E-1 Ref. Par	Test	Conditions	AQL (percent defective)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	GENERAL							
3.1	Qualification	Required;(Note 3)	---	---	---	---	---	---
3.6	Performance	(Note 4)	---	---	---	---	---	---
3.7	Ident. Marking	(Note 5)	---	---	---	---	---	---
4.9.2.1	Dimensions	See Fig. 1 herein	---	---	---	---	---	---
4.6.1	Preheating	---	---	---	---	---	---	---
4.5	Holding period	t = 48 hrs., min	---	---	---	---	---	---
4.9.18.1.2	Container Drop	Required;(Note 6)	---	---	---	---	---	---

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MIL-E-1 Ref. Par.	Test	Conditions	AQL (percent defective)	Insp Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	QUALIFICATION INSPECTION- (NOTE 6)							
4.12.4.1	Cathode illumination	---	---	---	---	---	---	---
4.12.3.4	Alignment, neck and bulb	---	---	---	---	---	0.25	in.
4.10.14	Capacitance (interelectrode)	g1 to all	---	---	C	---	7.0	uuf
		k to all	---	---	C	---	6.0	uuf
		D1 to D2	---	---	C	---	4.0	uuf
		D3 to D4	---	---	C	---	3.5	uuf
		D1 to all	---	---	C	---	12.0	uuf
		D2 to all	---	---	C	---	12.0	uuf
		D3 to all	---	---	C	---	7.0	uuf
		D4 to all	---	---	C	---	7.0	uuf
4.9.11	Pressure	---	---	---	---	---	---	---
4.9.19.8	Vibration	---	---	---	Width	---	1.0	mm
4.12.12	Deflection-factor uniformity	---	---	---	diff.	---	2.5	%
	ACCEPTANCE INSPECTION, PART 1 (PRODUCTION)- (NOTE 7)							
4.12.1.1	Cathode current	Ib3 = 25 uAdc (Note 8)	---	---	Ik	---	1000	uAdc
4.12.1.2	Voltage breakdown	---	---	---	---	---	---	---
4.12.1.3	Voltage breakdown	---	---	---	---	---	---	---
4.12.2.1	Gas "cross"	Ib3=5 uAdc (Note 8)	---	---	---	---	---	---
4.12.3.7	Angle between traces	---	---	---	---	89	91	Degr.
4.12.5.1	Blemishes	---	---	---	---	---	---	---
4.12.5.3	Modulation	Ib3=2 uAdc (Note 8)	---	---	ΔEc	---	18	Vdc

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MIL-E-1 Ref. Par.	Test	Conditions	AQL (percent defective)	Insp Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	ACCEPTANCE INSPECTION, PART 1 (PRODUCTION)-(NOTE 7)-(Continued)							
4.12.7.2	Spot position	---	---	---	---	---	20	mm
4.12.7.3	Spot displacement (leakage)	---	---	---	displ.	---	10	mm
4.12.9	Grid cutoff voltage	---	---	---	Eco	-135	-202	Vdc
4.12.10.1	Focusing voltage at cutoff	---	---	---	Eb1	---	2225	Vdc
4.12.10.2	Focusing voltage, zero-bias	---	---	---	Eb1	1510	---	Vdc
4.12.13.2	Grid No.1 leakage	---	---	---	---	---	---	---
4.12.13.4	Anode No.1 leakage	---	---	---	---	---	---	---
4.12.13.5	Anode No.2 leakage	---	---	---	---	---	---	---
---	Pattern distortion	(Note 9)	---	---	---	---	---	
	ACCEPTANCE INSPECTION, PART 2 (DESIGN)							
4.9.5	Base, cap, and insert secureness	---	6.5	L6	---	---	---	---
3	Heater current	---	6.5	L6	If	540	660	mA
4.12.1.1	Anode No.1 current	1b3 = 25 uAdc (Note 8)	6.5	L6	Ib1	-15	10	uAdc

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MIL-E-1 Ref. Par.	Test	Conditions	AQL (percent defective)	Insp Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	ACCEPTANCE INSPECTION, PART 2 (DESIGN)-(Continued)							
4.12.3.1	Alignment, base	+1 D2; Pin No. 1	6.5	L6	---	---	---	---
4.12.3.2	Alignment, side terminal	+1D2	6.5	L6	---	---	---	---
4.12.3.5	Alignment, base-and-neck	---	6.5	L6	---	---	---	---
4.12.3.3	Alignment, side terminal and base	Pin No. 1	6.5	L6	---	---	---	---
4.12.4.2	Stray emission	Eb2 = 6700 Vdc Eb3 = 11,000 Vdc	6.5	L6	---	---	---	---
4.12.5.4	Screen (P7)	---	6.5	L6	---	---	---	---
---	Screen (P19)	(Note 10)	6.5	L6	cbf	575	---	cb
					D 0.1	410	---	cb
					D 1.0	335	---	cb
					D 1.0-D10.0	75	120	cb
4.12.6.1	Line width:	Ib3 = 2 uAdc (Note 11)	6.5	L6	---	---	---	---
	Aa				width	---	0.32	mm
	Ab				width	---	0.75	mm
	Ac				width	---	0.75	mm
	Ad				width	---	0.32	mm
	Ae				width	---	0.32	mm
	Ba				width	---	0.32	mm
	Bb				width	---	0.60	mm
	Bc				width	---	0.60	mm
	Bd				width	---	0.75	mm
	Be				width	---	0.75	mm
4.12.11	Deflection factor	D1D2	6.5	L6	DF	100	150	Vdc/in.
4.12.11	Deflection factor	D3D4	6.5	L6	DF	100	150	Vdc/in.
4.12.13.1	Heater-cathode leakage	---	6.5	L6	---	---	---	---

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MIL-E-1 Ref. Par.	Test	Conditions	AQL (percent defective)	Insp Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	ACCEPTANCE INSPECTION, PART 3, (LIFE)							
4.11.2	Life test	Group C t - 500 hrs.	---	---	---	---	---	---
4.11.1.2	Cathode-ray tubes	Ib3 - 2 uAdc Eb2 = 6700 Vdc Eb3 = 11,000 Vdc	---	---	---	---	---	---
4.11.4	Life-test end points:	Ib3 = 2 uAdc	---	---	---	---	---	---
	Line width Aa	(Note 11)	---	---	width	---	0.36	mm
	Line width Ba	(Note 11)	---	---	width	---	0.36	mm
	Modulation	---	---	---	ΔEc	---	21	Vdc
5	Preparation for delivery (Note 12)		---	---	---	---	---	---

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NOTES:

1. Refer to Specification MIL-E-1, paragraph 6.5.

2. The deflection-electrode circuit resistances should be approximately equal. Higher resistance values: up to 5.0 meg, should be used for low beam-current operation.

3. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-1, supplement (Army), whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Standardization Engineering Division, U. S. Army Electronics Materiel Support Agency, Fort Monmouth, New Jersey, attention: SELMS-PSM-3.

4. Associable with and in addition to the performance requirements covered herein, all tests listed in Specification MIL-E-1, paragraph 3.6, are applicable except those per reference paragraphs 4.7, 4.9.1.1, 4.9.3, 4.9.5.1, 4.9.8, 4.9.20.1, 4.9.20,2, and 60.1 (App'x B).

5. Type-designation marking of electron tubes procured on Department of Army contracts, and which have passed Government inspection and comply with all requirements of this specification, shall consist of: "USA-manufacturer's qualification code letters-tube designation." The letters "JAN" or any abbreviation thereof shall not be used. If any specification waiver has been granted, the combination "USA-manufacturer's qualification code letters" shall not be used to complete the type-designation marking.

6. All tests listed hereon, except Container Drop, shall be performed during qualification inspection; however, the six tests listed herein under "Qualification Inspection" normally are performed during qualification inspection only.

7. The AQL for the combined defectives for attributes, excluding Inoperatives and Mechanical, shall be 1.0%. A tube having one or more defects shall be counted as one defective. Standard MIL-STD-105, Inspection Level II, shall apply.

8. The P19 screen can be permanently damaged if the current density is permitted to rise too high; to prevent burning, minimum beam-current densities should be employed. Tests requiring high beam currents may be made by measuring beam current at the deflection plates. To make this measurement, deflect a 1 x 1-inch low-intensity raster in the D1-D2 direction until the raster cannot be seen on the screen; then, readjust Ec1 to give the desired current.

9. The edges of a 7.5 x 7.5-inch raster pattern centered on the tube face shall fall completely within the boundaries of an identically-centered 7.625-in. square and an identically-centered inscribed 7.375-in. square.

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NOTES:

10. Guidances in Specification MIL-E-1, paragraph 4.12.5.4 shall be applicable to the following particular test conditions for P19 screen test:

    Eb3 = 9700 Vdc; Eb2 = 6100 Vdc; Eb1 - focus; Ib3 - 2.0 uAdc; raster size = 71 x 71 mm; duration of raster = 1/60 sec.; raster repeated at 1-sec. intervals.

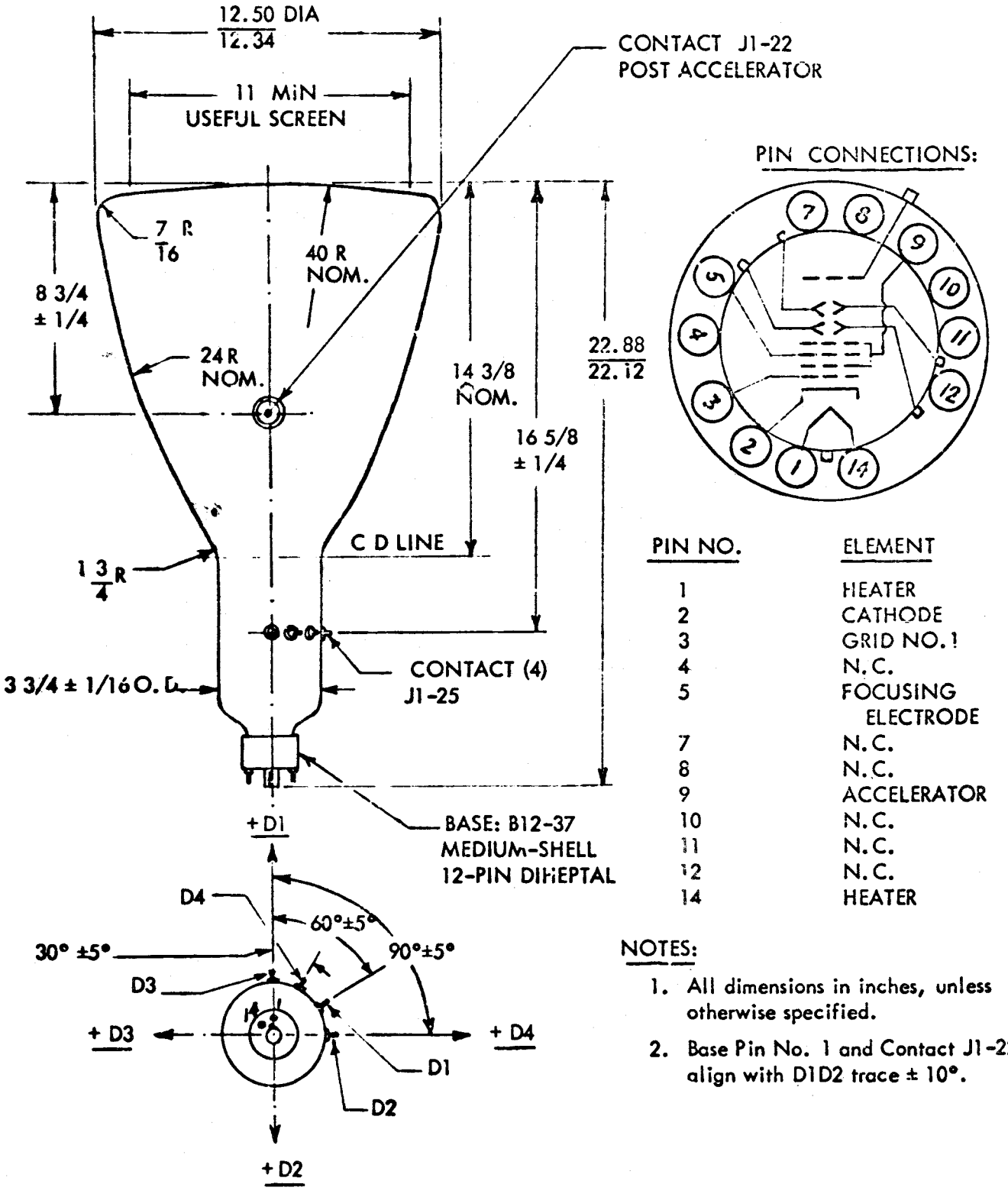
    cbf is fluorescent light output under continuous raster excitation. D0.1, D1.^, and D10.0 are, respectively, light outputs 0.1 second, 1.0 second, and 10.0 seconds after the excitation (tenth raster) is removed.

11. Line Width "An" is defined as the line width measured at a point "N" (a, b, c, d, e: see Fig. 2 herein) on the tube face, with the high-frequency scan applied to the D1D2 deflection plates and the 60-cycle sawtooth scan applied to the D3D4 plates. Line Width "Bn" is defined as the line width measured at the pertinent point "N" on the tube face with the plate connections interchanged and the focus setting unaltered.

    Refer to Fig. 2 herein for location of line-width measurement points. Using a 4-inch raster, focus for best line width Aa; without changing the focus or astigmatism setting, measure line widths Ab, Ac, Ad, and Ae; measurements shall not exceed the specified limits. Now, reversing the scan direction, measure line widths Ba, Bb, Bc, Bd, and Be, without changing the focus or astigmatism setting; measurements shall not exceed the specified limits.

12. Tubes shall be packaged and packed, as specified in the bid request or contract, in accordance with Specification MIL-E-75 and Appendix thereto.

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PIN CONNECTIONS:

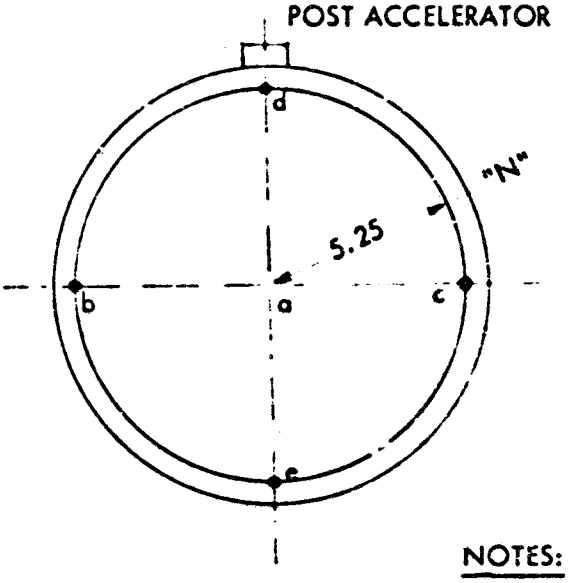
PIN NO.	ELEMENT
1	HEATER
2	CATHODE
3	GRID NO. 1
4	N. C.
5	FOCUSING ELECTRODE
7	N. C.
8	N. C.
9	ACCELERATOR
10	N. C.
11	N. C.
12	N. C.
14	HEATER

NOTES:

1. All dimensions in inches, unless otherwise specified.
2. Base Pin No. 1 and Contact J1-22 align with D1D2 trace ± 10°.

Figure 1. Outline and dimensions.

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NOTES:

1. All dimensions in inches.

Figure 2. Line-Width measurement points.

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